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NATURAL LANGUAGE PROCESSING ANALYSIS OF PUREGYM CUSTOMER REVIEWS

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1. INTRODUCTION

This analysis examines customer feedback from PureGym, a leading value fitness operator with 2 million members across 600 gyms. Using customer reviews data from Google Reviews (23,250) and Trustpilot (16,673), we employed multiple Natural Language Processing (NLP) techniques to analyse customer sentiment and identify improvement opportunities.

2. APPROACH AND INITIAL DATA ANALYSIS

Our methodology combined traditional NLP analysis, topic modelling (BERTopic), emotion analysis (BERT), large language model analysis (Phi-3.5), and Latent Dirichlet Allocation (LDA) to generate actionable insights for enhancing customer experience while maintaining PureGym's core value proposition of affordable, accessible fitness facilities.

After cleaning for missing comments, Google dataset reduced to 13,898 reviews, while Trustpilot maintained original size.

Multilingual preprocessing pipeline with language detection (EN/DA/DE), custom stopwords, and phrase preservation. Includes POS tagging, lemmatization, domain-specific term handling, and context-aware tokenization optimised for international gym reviews.

LOCATION AND WORD FREQUENCY ANALYSIS

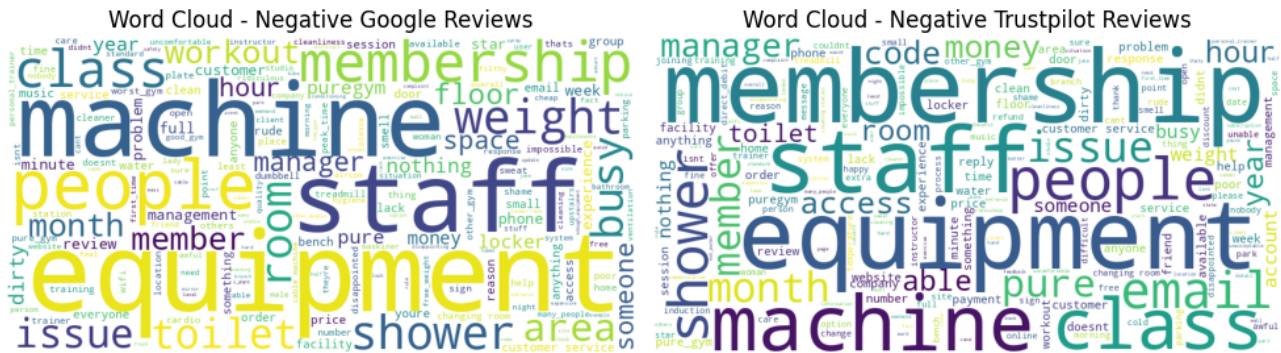
- Google reviews - 512 locations, Trustpilot 377 locations, common locations - 310.
- Common dominant terms across platforms: 'equipment' (Google: 2768, Trustpilot: 3354), 'class', 'great', 'staff'.
- Platform-specific emphasis: 'weight'/'busy' (Google) versus 'membership'/'easy' (Trustpilot).

Fig 1: Word Cloud - All Reviews (Google and Trustpilot)



- Negative reviews distribution: Google 20.0% (2,785), Trustpilot 21.2% (3,543)
- Primary Concerns: Equipment/machine issues (936/754 mentions), staff interactions (~650 mentions)
- Platform Focus: Google emphasises facilities ("equipment", "machine"), Trustpilot highlights services ("membership", "machine")

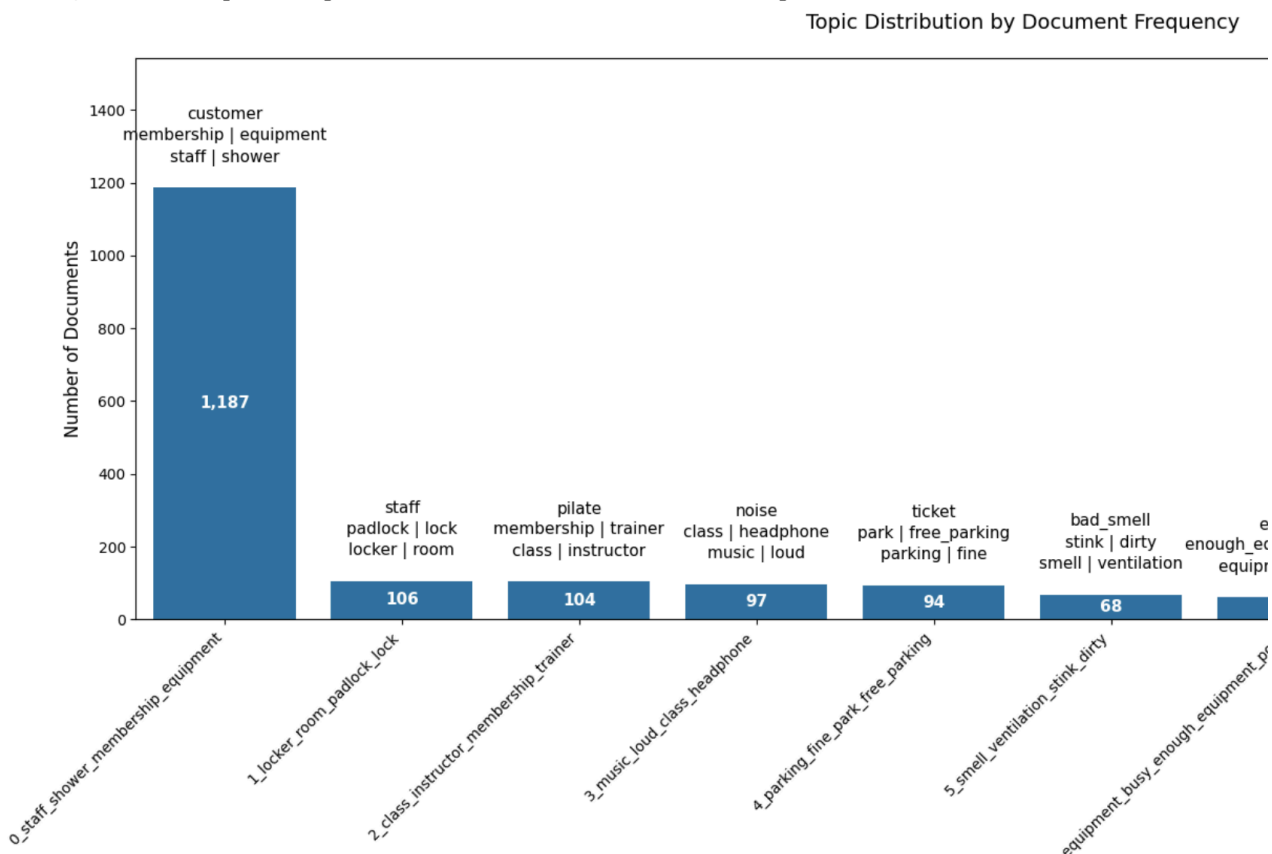
Fig 2: Word Cloud - Negative Reviews (Google and Trustpilot)



3. TOPIC MODELLING ANALYSIS

BERTopic analysis of negative reviews for all locations refined 53 initial topics to 28 key clusters, revealing distinct customer feedback patterns.

Fig 3: Top 5 topics of BERT analysis - negative reviews all locations



Analysis reveals key issues: (1) Basic facilities and membership management (1,187 reviews), (2) Staff interactions and locker access (326), (3) Class and trainer problems (104), (4) Noise control concerns (97), and (5) Parking system complaints (94).

4. FURTHER DATA INVESTIGATION WITH LOCATION FILTERING

LOCATION-BASED VARIATIONS FOR NEGATIVE REVIEWS

London dominates negative feedback with 8-11 locations in top 20 across both platforms. Key overlapping locations include Enfield, Stratford, Swiss Cottage, Bermondsey, Hayes, and Seven Sisters. London Stratford leads Google reviews (13.47%), while Leicester Walnut Street tops Trustpilot complaints (12.76%). Total negatives: Google (438), Trustpilot (392)

COMPARATIVE ANALYSIS OF WORD CLOUDS

Key differences between all-location and top-30 analyses reveal distinct patterns. While all-location reviews emphasise equipment and facilities (2700+ mentions), top-30 locations show stronger focus on class experience (700+ mentions) and service quality. Top-30 locations display more positive terms ('great_gym', 'friendly') and balanced feedback across operational aspects.

Notable Changes:

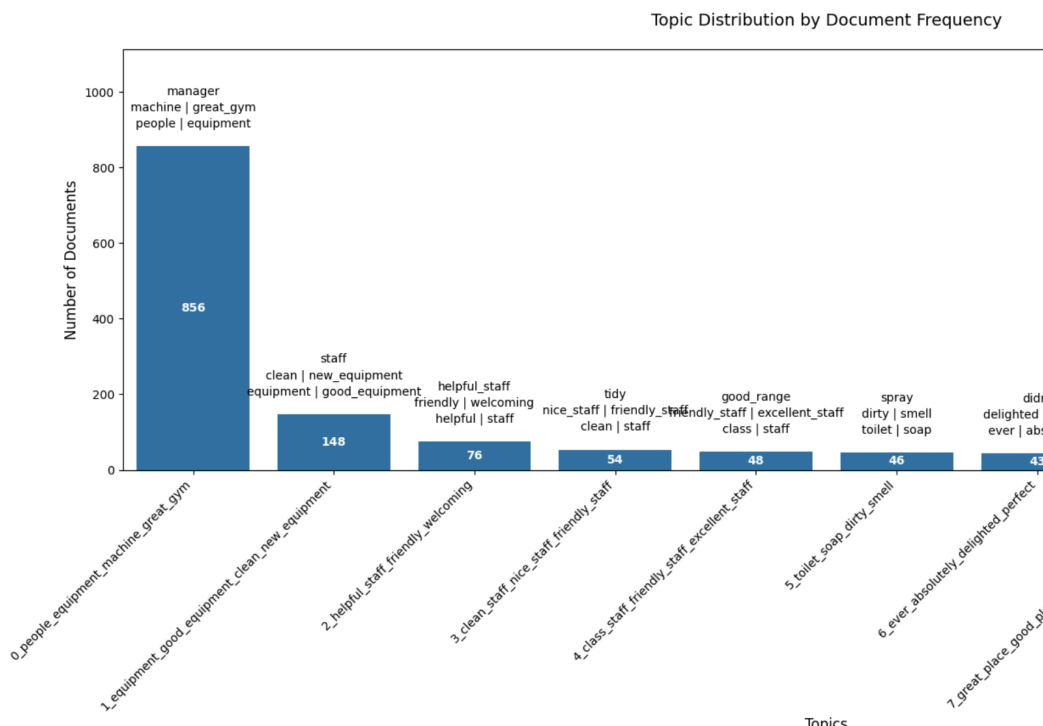
1. Word frequency scale (4x higher in all-locations)
2. Priority shift from equipment to class focus
3. More positive sentiment in top-30 locations
4. Better balanced distribution of terms in top-30

Fig.4: Top 30 locations All Reviews



BERTopic analysis of all reviews for top 30 locations refined 69 initial topics to 48 key clusters, revealing distinct customer feedback patterns.

Fig.5: Document frequencies from BERTopic top 30 locations All Reviews



The top clusters reveal: (1) Management & gym operations are the primary concern (856 reviews), (2) Facility maintenance and cleanliness (148), (3) Customer service quality (76), (4) Class experience and instruction (54), and (5) Staff professionalism and friendliness (48).

5. EMOTION ANALYSIS

Joy dominates both platforms (9,500+ Google, 11,700+ Trustpilot), followed by anger (2,400+ each). In negative reviews, anger leads (5,000), then sadness (2,500) and fear (800). Anger-related topics cluster around facility/equipment issues (1,690 reviews) and personnel/service problems (639).

Fig.6: Emotion distribution

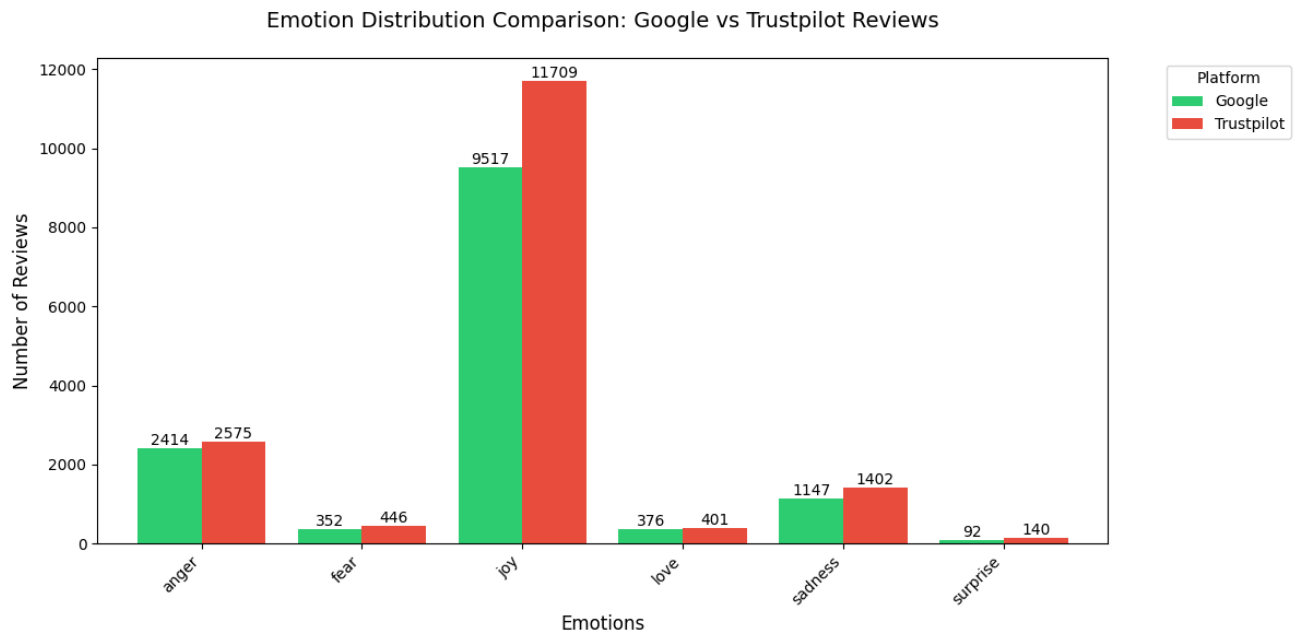
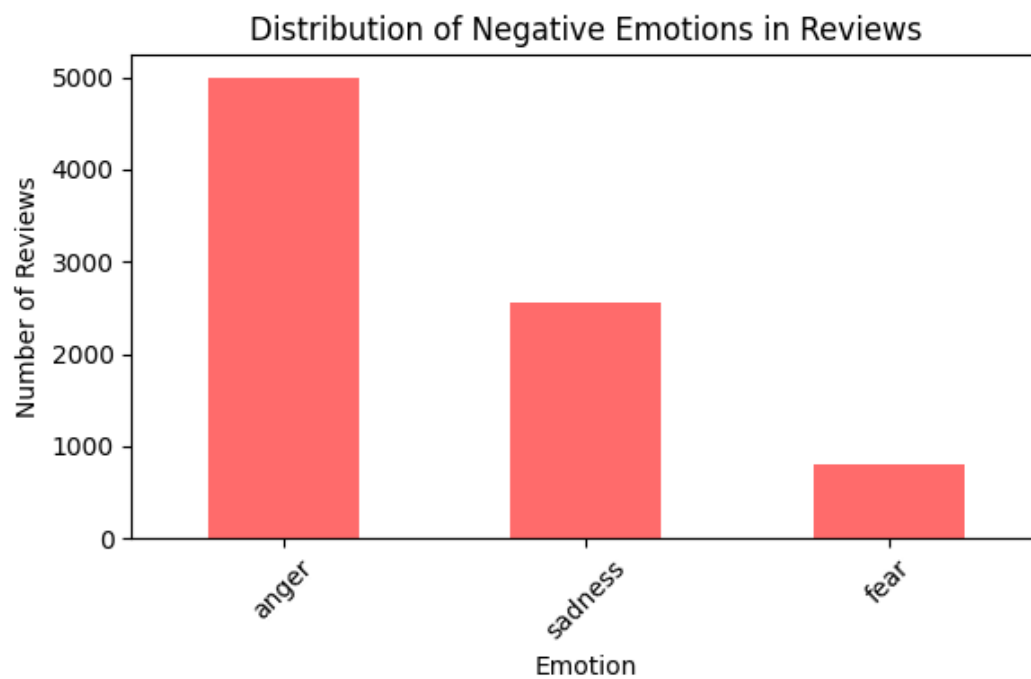


Fig.7: Negative emotion distribution



The topic modelling analysis of anger-related reviews reveals two major clusters:

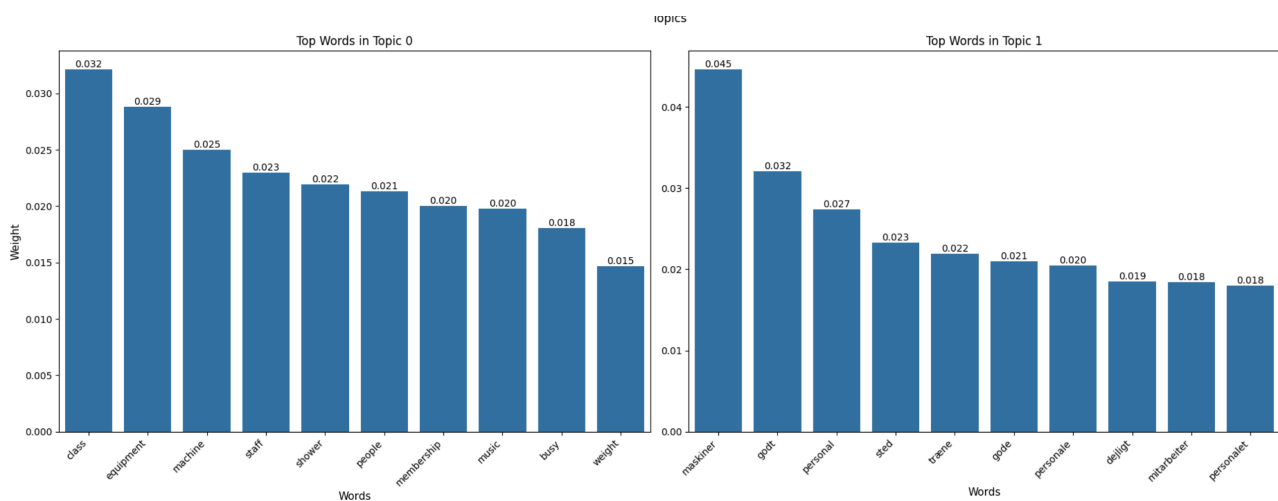
1. Facility and Equipment Issues (Topic 0, Size: 1,690 documents):

- Primary concerns center around "class", "equipment", "machine", "staff", and "shower" facilities
- Shows strong correlation with operational aspects of the gym

2. Personnel and Service Issues (Topic 1, Size: 639 documents):

- Focuses on "maskiner" (machines), "godt" (good), "personal", "sted" (place), and "træne" (train)
- Indicates issues related to staff interactions and training facilities

Fig.8: Top 2 anger topic top words



6. LANGUAGE MODEL FROM HUGGINGFACE

The LLM-enhanced BERTopic analysis reveals significantly more granular insights compared to previous analyses. The model identified 30 distinct topic clusters, providing a more comprehensive understanding of customer concerns.

Key Improvements:

1. Equipment and Maintenance (236 instances)

- Separated into general maintenance, quality issues, and availability problems
- Specific focus on cross-training equipment shortages
- Distinct categorisation of machine quality concerns

2. Service Quality (181 instances)

- Detailed breakdown of staff behaviour and response patterns
- Specific attention to instructor professionalism
- Customer service communication issues identified

3. Environmental Factors

- Detailed ventilation and temperature control issues
- Specific water facility problems (drinking fountains, showers)
- Space utilisation and capacity management

New Insights:

- Digital Infrastructure: Emerged as significant concern (app functionality, software issues)
- Class Management: More detailed view of scheduling and cancellation impacts
- Security: Specific focus on locker room and changing facility security

4. USING LDA MODEL FROM GENSIM

LDA analysis reveals several insights not captured in previous BERT or LLM analyses. Key new findings include the connection between temperature control and water systems, previously unidentified safety concerns, and the impact of email communication on service perception. The analysis also shows stronger relationships between seemingly unrelated issues like music and induction problems.

Unique discoveries:

1. Space utilisation impacts beyond equipment availability
2. Management response patterns to facility issues
3. Correlation between timing and complaint types
4. Email communication's role in membership satisfaction

5. FINAL RECOMMENDATIONS

Based on comprehensive analysis using BERTopic, LDA, and LLM-enhanced clustering, three critical areas require immediate attention:

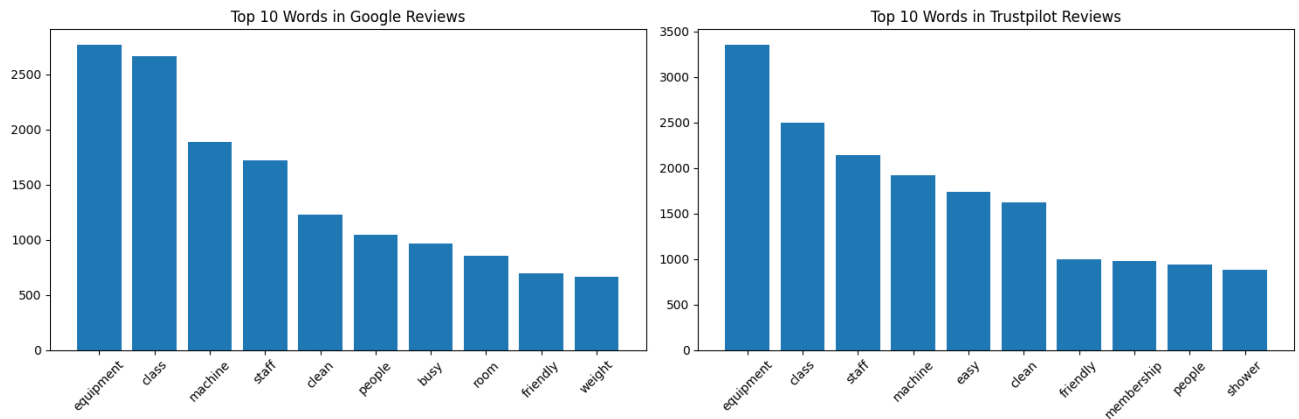
1. Equipment & Maintenance (Priority 1)
 - Implement preventive maintenance schedule
 - Add capacity during peak hours
 - Upgrade outdated machines
2. Staff & Service Excellence (Priority 2)
 - Enhance staff training for customer service
 - Improve response times for complaints
 - Standardise class instructor quality
3. Facility Management (Priority 3)
 - Upgrade ventilation/temperature control systems
 - Improve cleaning protocols
 - Enhance digital infrastructure (app/access systems)

ROI can be maximised by addressing these issues sequentially, focusing on equipment-related improvements first.

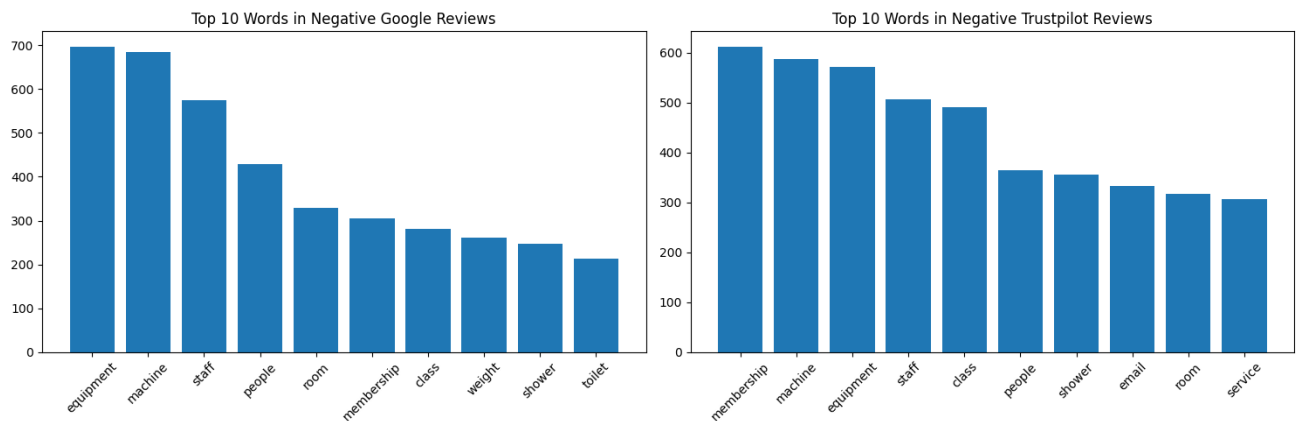
4. APPENDIX

4.1.A: EDA

A.1: Top 10 Words - All Reviews (Google and Trustpilot)



A.2: Top 10 Words - Negative Reviews (Google and Trustpilot)



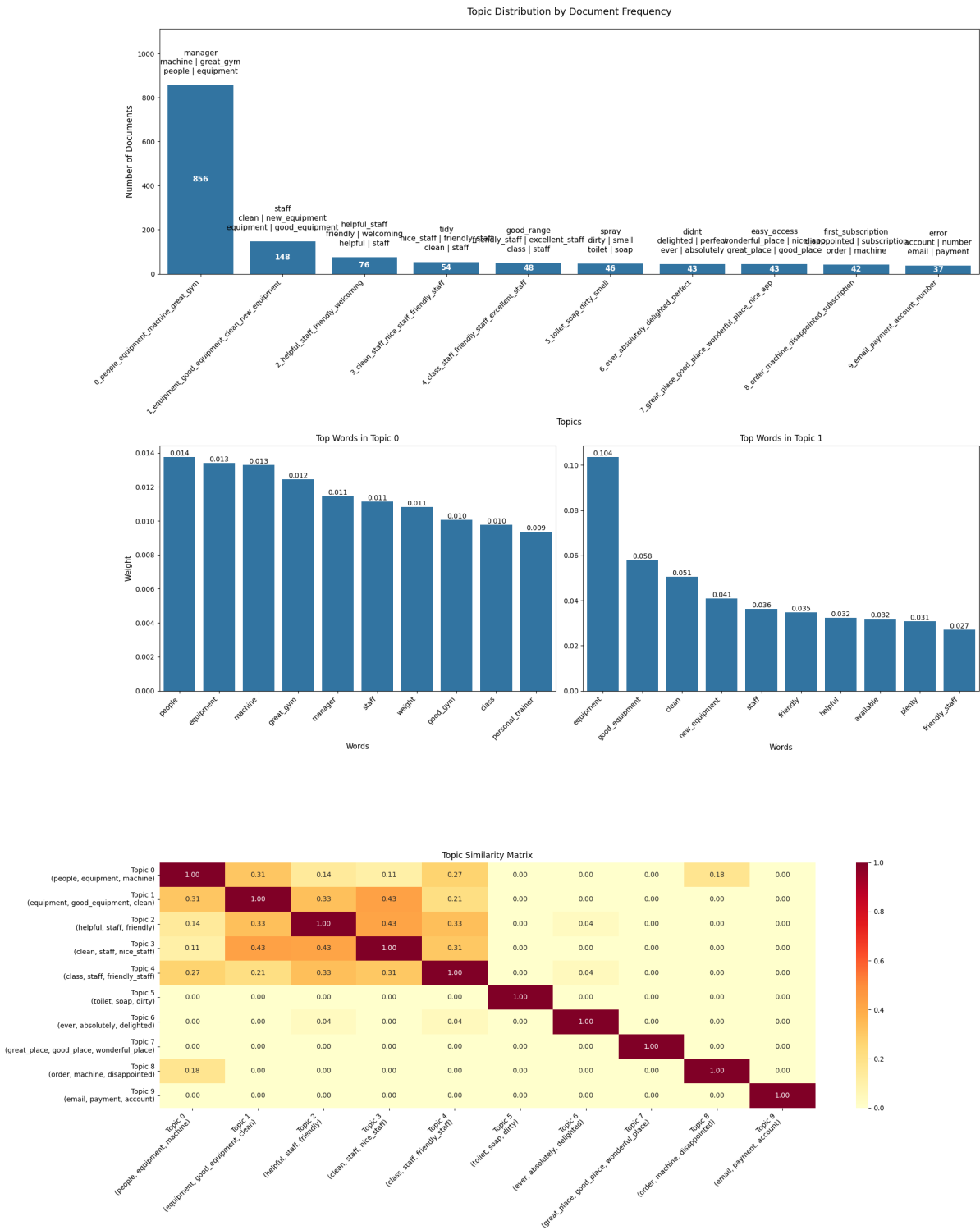
4.2.B: INITIAL TOPIC MODELLING

B.1: Document frequencies, top words & Similarity heatmap



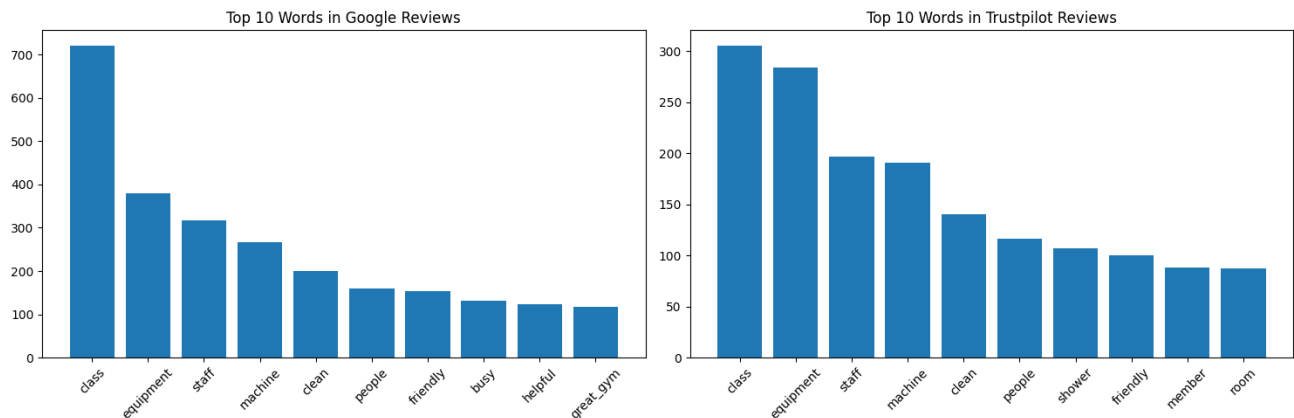
4.3.C: FILTERED LOCATIONS INVESTIGATION

C1: Top 30 locations negative reviews BERT analysis



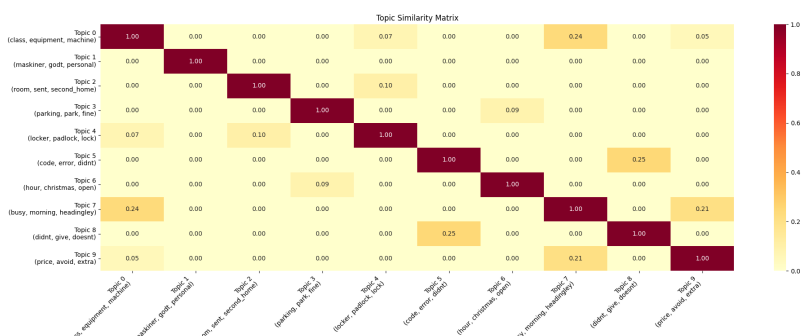
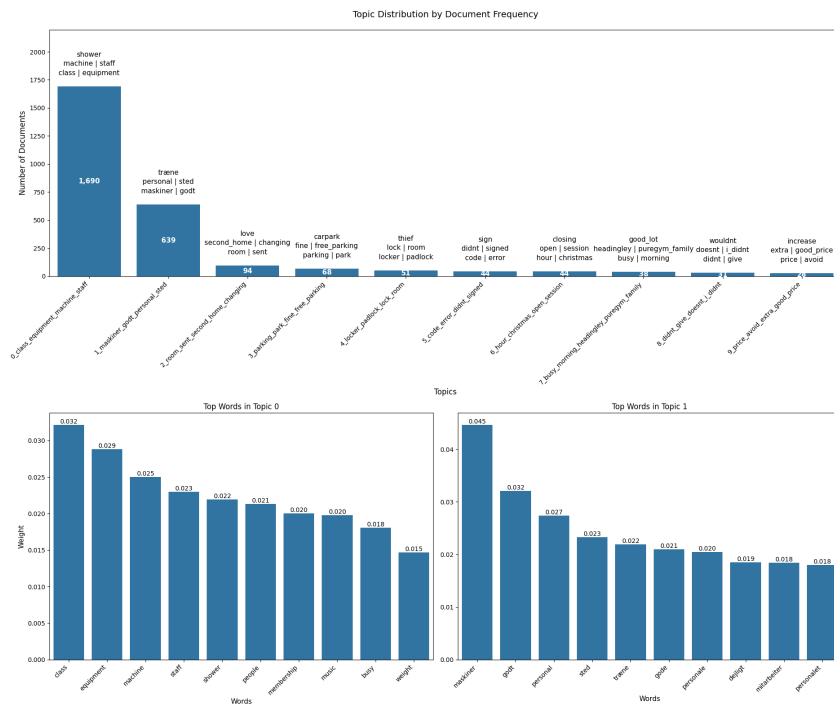
PREDICT STUDENT DROPOUT USING SUPERVISED AND DEEP LEARNING

C.2: Word frequencies Top 30 locations negative reviews

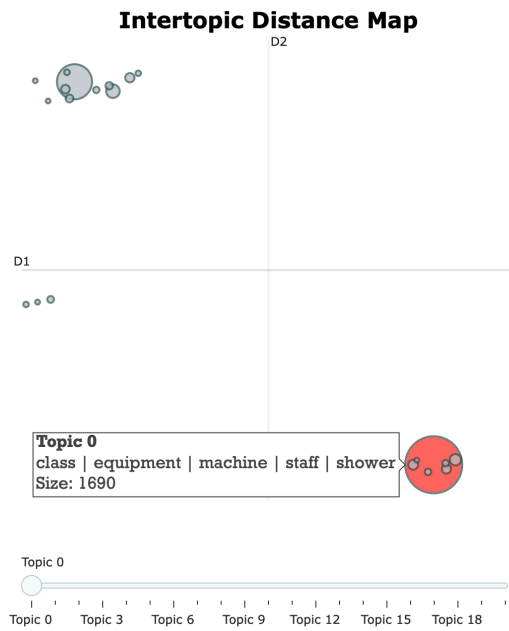


4.4.D: EMOTIONAL ANALYSIS

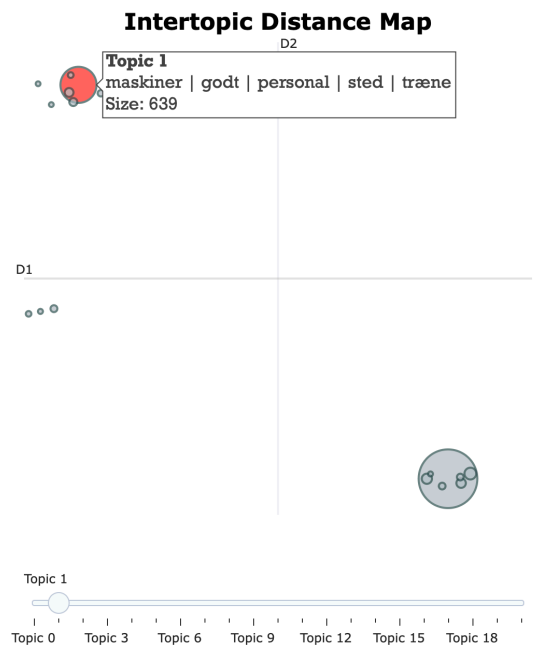
D1: Negative reviews with anger emotion BERT analysis



D.2: Intertopic distance map: Topic 0



D.3: Intertopic distance map: Topic 1



4.5.E: LLM + BERT ANALYSIS

E.1: initialisation of LLM Phi-3.5-mini-instruct for sample size 500 negative reviews

```
config.json: 100% 3.45k/3.45k [00:00<00:00, 307kB/s]
configuration_phi3.py: 100% 11.2k/11.2k [00:00<00:00, 930kB/s]
A new version of the following files was downloaded from https://huggingface.co/microsoft/Phi-3.5-mini-instruct:
- configuration_phi3.py
. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file, you can pin a revision.
modeling_phi3.py: 100% 73.8k/73.8k [00:00<00:00, 3.48MB/s]
A new version of the following files was downloaded from https://huggingface.co/microsoft/Phi-3.5-mini-instruct:
- modeling_phi3.py
. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file, you can pin a revision.
model.safetensors.index.json: 100% 16.3k/16.3k [00:00<00:00, 1.20MB/s]
Downloading shards: 100% 2/2 [03:02<00:00, 86.42s/it]
model-00001-of-00002.safetensors: 100% 4.97G/4.97G [01:59<00:00, 42.0MB/s]
model-00002-of-00002.safetensors: 100% 2.67G/2.67G [01:03<00:00, 42.7MB/s]
Loading checkpoint shards: 100% 2/2 [00:03<00:00, 1.72s/it]
generation_config.json: 100% 195/195 [00:00<00:00, 16.9kB/s]
tokenizer_config.json: 100% 3.98k/3.98k [00:00<00:00, 382kB/s]
tokenizer.model: 100% 500k/500k [00:00<00:00, 38.0MB/s]
tokenizer.json: 100% 1.84M/1.84M [00:00<00:00, 19.5MB/s]
added_tokens.json: 100% 306/306 [00:00<00:00, 25.8kB/s]
special_tokens_map.json: 100% 665/665 [00:00<00:00, 53.3kB/s]
The `seen_tokens` attribute is deprecated and will be removed in v4.41. Use the `cache_position` model input instead.
`get_max_cache()` is deprecated for all Cache classes. Use `get_max_cache_shape()` instead. Calling `get_max_cache()` will raise error from v4.48
LLM setup complete!
Processing 500 negative reviews
Debug - Starting to process 500 reviews in 50 batches
```

PREDICT STUDENT DROPOUT USING SUPERVISED AND DEEP LEARNING

E.2: Feeding LLM Output to BERT and getting BERT outputted topics

```
Debug - Final total topics: 1517
Debug - Sample of final topics:
- Overcrowding and group management
- Exercise engagement
- Equipment usage and governance
- Cleanliness and hygiene issues
- Lack of water supply and facilities
```

Extracted 1517 topics total

2024-11-23 16:37:54,727 - BERTopic - Embedding - Transforming documents to embeddings.

Batches: 100%  48/48 [00:00<00:00, 127.63it/s]

```
2024-11-23 16:37:55,639 - BERTopic - Embedding - Completed ✓
2024-11-23 16:37:55,640 - BERTopic - Dimensionality - Fitting the dimensionality reduction algorithm
2024-11-23 16:38:00,471 - BERTopic - Dimensionality - Completed ✓
2024-11-23 16:38:00,473 - BERTopic - Cluster - Start clustering the reduced embeddings
2024-11-23 16:38:00,676 - BERTopic - Cluster - Completed ✓
2024-11-23 16:38:00,677 - BERTopic - Representation - Extracting topics from clusters using representation models.
2024-11-23 16:38:00,743 - BERTopic - Representation - Completed ✓
2024-11-23 16:38:00,744 - BERTopic - Topic reduction - Reducing number of topics
2024-11-23 16:38:00,785 - BERTopic - Topic reduction - Reduced number of topics from 48 to 32
```

Top Topics Identified:

Topic 0: 0_equipment_maintenance_quality_overall (Count: 236)

Key terms: equipment, maintenance, quality, overall, machines

Topic 1: 1_staff_customer_response_behavior (Count: 181)

Key terms: staff, customer, response, behavior, service

Topic 2: 2_cleanliness_hygiene_cleaning_sanitation (Count: 115)

Key terms: cleanliness, hygiene, cleaning, sanitation, maintenance

Topic 3: 3_unzureichende_maskiner_mangel_kundenservice (Count: 54)

Key terms: unzureichende, maskiner, mangel, kundenservice, online

Topic 4: 4_membership_process_access_difficulty (Count: 38)

Key terms: membership, process, access, difficulty, upgrade

Debug - Generating insights from BERTopic analysis...

Actionable Insights Based on Topic Analysis:

4.6.F: LDA

F.1: LDA Visualisation

